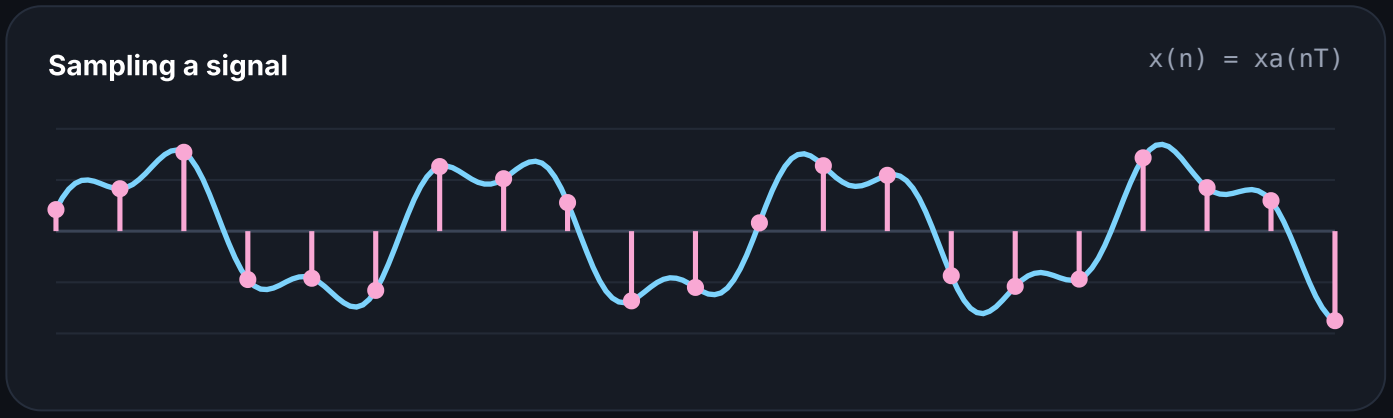
النظري
Chapter 1

Digital Signal Processing · Theory · 22 questions



Signals & Systems | Frequency | A/D & D/A | Sampling | Aliasing | Quantization

01	محاضرة 2	7
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Lecturer: Dr. Samir M. Kamal

Modern Academy for Engineering & Technology

DSP / ch01 / contents

المحتوى

Theory · 22 questions

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5 سؤال p. 14

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A/D and D/A Conversion
10 سؤال p. 22

إزاي تذاكر من الملف

الإجابات بجُمل الدكتور. جوه كل محاضرة الأسئلة مترتبة من الأكتر تكرارًا في الامتحانات. المربع الأخضر (الفكرة) هو الإجابة في سطر، و Key words تسمّع منها وانت مغطي الإجابة. في آخر كل محاضرة Cheat sheet.

DSP / ch01 / repeated

الأسئلة النظري اللي بتتكرر أكتر (ابدأ بيها)

Most repeated in previous exams — start with these

السؤال	عدد مرات وروده
Q9. In detail, what are the main characteristics of the discrete-time sinusoidal signals?	9
Q18. In A/D, explain in detail the quantization process of continuous-amplitude signals.	8
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Q16. In sampling process, derive the sampling condition $Fs \geq 2f_{max}$ (avoiding aliasing).	6
Q22. Check (✓) for the right statements and (×) for the wrong statements and correct the mistakes.	4
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Q13. Explain (with a block diagram) the basic three operations performed in the A/D converter.	4
Q8. In detail, what are the main characteristics of the continuous-time sinusoidal signals?	4
Q4. Explain, with drawings, the different classifications of signals.	4

SECTION 01

محاضرة 2

Signals, Systems & Classification

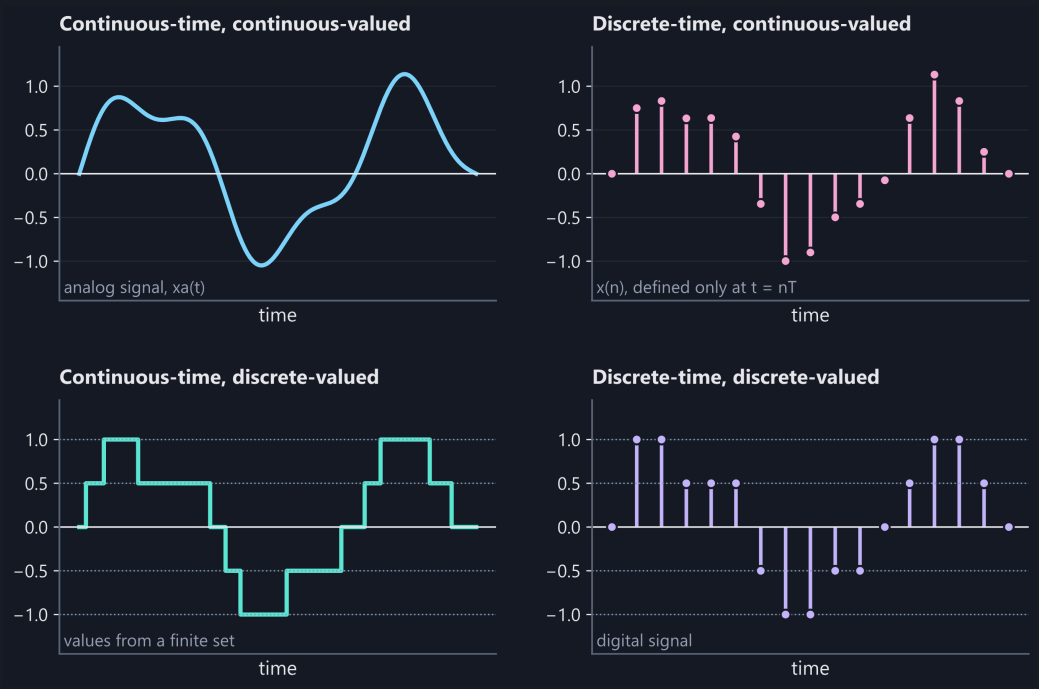
الأسئلة — الأكثر تكرارًا الأول

p. 5	Explain, with drawings, the different classifications of signals.	Q4 ☐
p. 7	Define the following items: Signal, System, and Signal Processing (give an example f...	Q1 ☐
p. 9	Draw the block diagram of (a) an analog signal processor and (b) a digital signal pr...	Q2 ☐
p. 10	What are the advantages of digital signal processing over analog signal processing? ...	Q3 ☐
p. 11	Compare between deterministic and random signals.	Q5 ☐
p. 11	What is a digital signal? Why must a signal be digital to be processed digitally?	Q6 ☐
p. 12	What is quantization? Compare between truncation and rounding with an example.	Q7 ☐

Q4 متكرر ●4×

Explain, with drawings, the different classifications of signals.

1. **Continuous-time vs. discrete-time:** continuous-time (analog) signals are defined for every value of time and take on values in the continuous interval (a, b). Discrete-time signals are defined only at certain specific values of time.
2. **Continuous-valued vs. discrete-valued:** a continuous-valued signal takes on all possible values on a finite or an infinite range. A discrete-valued signal takes on values from a finite set of possible values.
3. **Deterministic vs. random:** a deterministic signal can be uniquely described by an explicit mathematical expression, a table of data, or a well-defined rule. Otherwise, the signal is random.
4. **Multichannel vs. multidimensional:** a multichannel signal is generated by multiple sources or sensors and represented in vector form, $S3(t) = [s1(t)\ s2(t)\ s3(t)]^T$. A signal is M-dimensional if its value is a function of M independent variables.



الفكرة Continuous vs discrete time, continuous vs discrete valued, deterministic vs random, multichannel vs multidimensional.

محاضرة 2

Key words Deterministic vs. random

المصدر — من الـ PDFs بتاعتك

المحاضرة **LEC 2** · Sec. 1.4 · p. 18
1-Lectures/Lectures Fall 2026/Chapter 1/DSP- LEC- 2- Ch-1.pdf

جه في Midterm 8-12-2020 · Q2-a | Revision Sheet 1 | Sheet 1 · 1.3 | نظري الميد
2-Section/Revision Sheet (summer)/ · 2-Section/Sheets/ · 5-Exams/Midterm/ · 6-Extra/

×3 Q1

Define the following items: Signal, System, and Signal Processing (give an example for each).

- **Signal:** is defined as any physical quantity that varies with time, space, or any other independent variable(s). Mathematically, we describe a signal as a function of one or more independent variables.
  Examples: $s1(t) = 5t$ (varies linearly with t), $s2(t) = 20t^2$ (varies quadratically with t), and $s(x, y) = 3x + 2xy + 10y^2$ (two independent variables, e.g. the spatial coordinates in a plane). A speech signal cannot be described by such an expression; a segment of it is represented as a sum of several sinusoids of different amplitudes, frequencies and phases.
- **System:** is defined as a physical device that performs an operation on a signal.
  Example: a filter used to reduce the noise and interference corrupting a desired information. If the operation is linear, the system is called linear; if it is nonlinear, the system is nonlinear.
- **Signal processing:** when we pass a signal through a system, as in filtering, we say that we have processed the signal.
  Example: filtering the noise and interference from the desired signal. Digital processing may be performed by digital hardware (logic circuits), by software, or by a combination of both.

الفكرة

Signal = physical quantity varying with an independent variable; system = device that operates on a signal; processing = passing a signal through a system (e.g. filtering).

Key words: Signal | System | Signal processing

المصدر — من الـ PDFs بتاعتك

المحاضرة

LEC 2 · Sec. 1.1 · p. 4
`1-Lectures/Lectures Fall 2026/Chapter 1/DSP- LEC- 2- Ch-1.pdf`

محاضرة 2

جه في

Sheet 1 · 1.1 | Midterm OLD · Model-C | Quiz 1 · Section

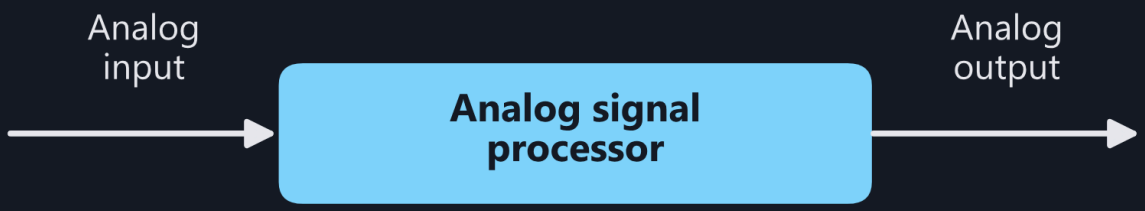
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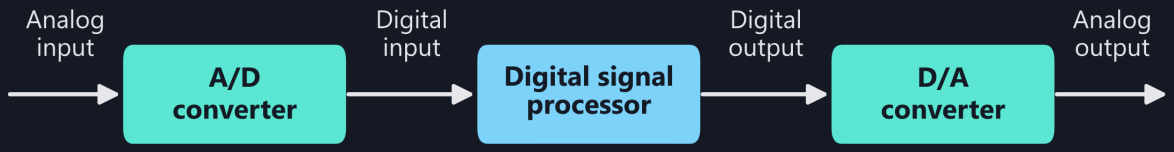
×2 Q2

Draw the block diagram of (a) an analog signal processor and (b) a digital signal processor.

There are two types of signal processing systems (Sec. 1.2):

1. **Analog signal processor (Fig. 1.1):** the analog input signal is processed directly by an analog signal processor to give an analog output signal.
2. **Digital signal processor (Fig. 1.2):** provides an alternative method for processing the analog signal. The **A/D converter** converts the analog input into a digital signal (a sequence of numbers), the **digital signal processor** performs the required operations on it, and the **D/A converter** converts the result back into an analog output signal.





الفكرة Analog: input → analog signal processor → output. Digital: input → A/D → digital signal processor → D/A → output.

المصدر — من الـ PDFs بتاعتك

المحاضرة **LEC 2** · Sec. 1.2 · p. 11

1-Lectures/Lectures Fall 2026/Chapter 1/DSP- LEC- 2- Ch-1.pdf

جه في Revision Sheet 1 · Sheet 1 · 1.2

2-Section/Revision Sheet (summer)/ · 2-Section/Sheets/

×1 Q3

What are the advantages of digital signal processing over analog signal processing? What is its main limitation?

1. **Flexibility:** a digital programmable system allows reconfiguring the operations simply by changing the program; an analog system needs hardware redesign.
2. **Accuracy:** a digital system provides much better control of accuracy requirements; tolerances in analog components make accuracy hard to control.
3. **Storage:** digital signals are easily stored on magnetic media without deterioration, so they are transportable and can be processed off-line.
4. Allows the implementation of more **sophisticated** signal processing algorithms.
5. **Precise mathematical operations** are very difficult in analog form but routine on a digital computer using software.
6. In some cases a digital implementation is **cheaper** than its analog counterpart.

Limitation: the speed of operation of A/D converters and digital signal processors; signals with extremely wide bandwidths may be beyond the state of the art of digital hardware.

الفكرة DSP wins in flexibility, accuracy, storage, sophisticated algorithms and cost; its limitation is the speed of the A/D converters and processors (bandwidth).

Key words Flexibility Accuracy Storage

المصدر — من الـ PDFs بتاعتك

المحاضرة **LEC 2** · Sec. 1.3 · p. 13
1-Lectures/Lectures Fall 2026/Chapter 1/DSP- LEC- 2- Ch-1.pdf

جه في Midterm OLD
5-Exams/Midterm/

Q5 ×1

Compare between deterministic and random signals.

- **Deterministic:** any signal that can be uniquely described by an explicit mathematical expression, a table of data, or a well-defined rule. All past, present, and future values are known precisely, without any uncertainty.
- **Random:** otherwise, we refer to these signals as random.

الفكرة

Deterministic = described exactly by an expression, table or rule (past, present and future known); random = cannot be predicted exactly.

Key words Deterministic Random

المصدر — من الـ PDFs بتاعتك

المحاضرة

LEC 2 · Sec. 1.4.3 · p. 26

1-Lectures/Lectures Fall 2026/Chapter 1/DSP- LEC- 2- Ch-1.pdf

جه في

Midterm OLD · Model-A 2013/2014

5-Exams/Midterm/

Q6 أقل تكرارًا

What is a digital signal? Why must a signal be digital to be processed digitally?

- A **digital signal** is discrete in time and its values are discrete (Fig. 1.4: a digital signal with four different amplitude values).
- In order for a signal to be processed digitally, it must be discrete in time and its values must be discrete (i.e., it must be a digital signal).

محاضرة 2

الفكرة

A digital signal is discrete in time and in amplitude; a digital processor can only handle sampled and quantized values.

المصدر — من الـ PDFs بتاعتك

المحاضرة

LEC 2 · Sec. 1.4.2 · p. 23

1-Lectures/Lectures Fall 2026/Chapter 1/DSP- LEC- 2- Ch-1.pdf

Q7 أقل تكرارًا

What is quantization? Compare between truncation and rounding with an example.

- **Quantization:** the process of converting a continuous-valued signal into a discrete-valued signal; it is basically an approximation process, accomplished by rounding or truncation.
- **Example:** if the allowable values are the integers 0 through 15, the value 8.58 is approximated by **8** by truncation, or by **9** by rounding to the nearest integer.

الفكرة

Quantization maps a continuous value to a finite set of levels; truncation drops the extra digits, rounding takes the nearest level.

Key words Quantization

المصدر — من الـ PDFs بتاعتك

المحاضرة

LEC 2 · Sec. 1.4.2 (truncation / rounding) · p. 25

1-Lectures/Lectures Fall 2026/Chapter 1/DSP- LEC- 2- Ch-1.pdf

المحاضرة

LEC 4 · Sec. 1.6.3 · p. 23

1-Lectures/Lectures Fall 2026/Chapter 1/DSP- LEC- 4- Ch- 1.pdf

محاضرة 2

★ Cheat sheet · محاضرة 2

افتكر

Signal = physical quantity varying with an independent variable; system = device that operates on a signal; processing = passing a signal through a system (e.g. filtering).	Q1
Analog: input → analog signal processor → output. Digital: input → A/D → digital signal processor → D/A → output.	Q2
DSP wins in flexibility, accuracy, storage, sophisticated algorithms and cost; its limitation is the speed of the A/D converters and processors (bandwidth).	Q3
Continuous vs discrete time, continuous vs discrete valued, deterministic vs random, multichannel vs multidimensional.	Q4
Deterministic = described exactly by an expression, table or rule (past, present and future known); random = cannot be predicted exactly.	Q5
A digital signal is discrete in time and in amplitude; a digital processor can only handle sampled and quantized values.	Q6
Quantization maps a continuous value to a finite set of levels; truncation drops the extra digits, rounding takes the nearest level.	Q7

SECTION 02

محاضرة 3

Concept of Frequency

الأسئلة — الأكثر تكرارًا الأول

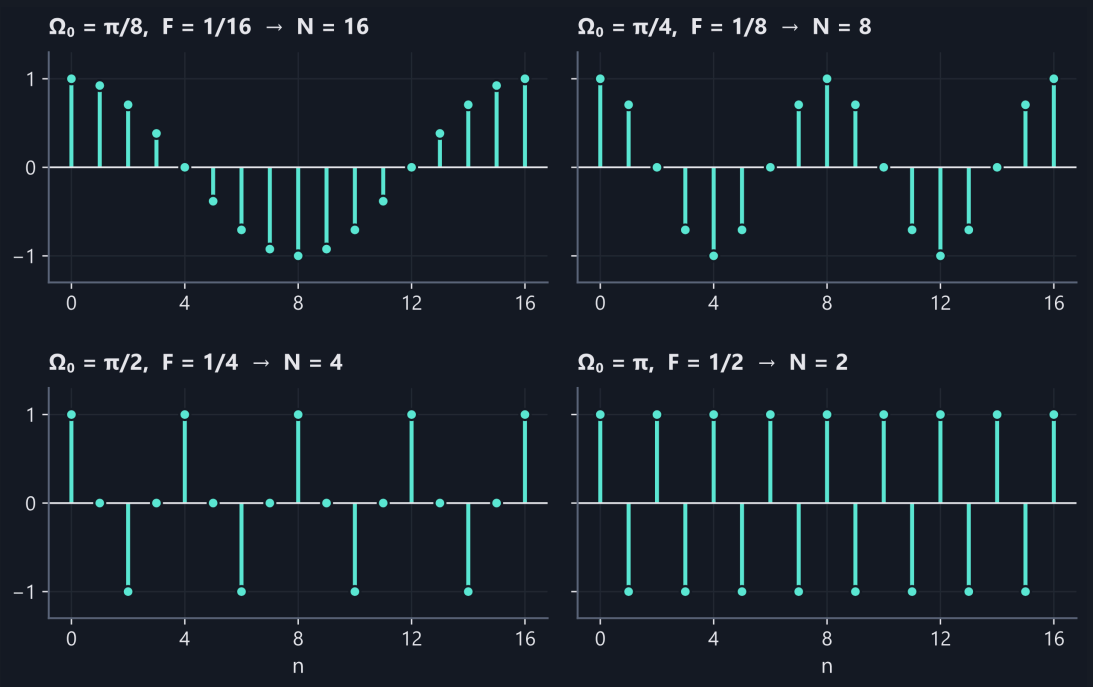
p. 15	In detail, what are the main characteristics of the discrete-time sinusoidal signals...	Q9 ☐
p. 17	In detail, what are the main characteristics of the continuous-time sinusoidal signa...	Q8 ☐
p. 18	Show that a discrete-time sinusoid is periodic only if its frequency F is a rational...	Q10 ☐
p. 19	Prove that discrete-time sinusoids whose frequencies are separated by an integer mul...	Q11 ☐
p. 20	What are harmonically related complex exponentials? Compare between the continuous-t...	Q12 ☐

Q9 متكرر ×9

In detail, what are the main characteristics of the discrete-time sinusoidal signals?

$x(n) = A\cos(\Omega n + \theta) = A\cos(2\pi F n + \theta)$, where n is the sample number, Ω is in radians/sample and F in cycles/sample.

1. **(a)** A discrete-time sinusoid is **periodic only if its frequency F is a rational number** ($F_o = k/N$).
2. **(b)** Discrete-time sinusoids whose frequencies are separated by an integer multiple of 2π are **identical**. Frequencies in $-\pi \le \Omega \le \pi$ ($-1/2 \le F \le 1/2$) are unique; all frequencies $|\Omega| > \pi$ are **aliases**.
3. **(c)** The **highest rate of oscillation** is attained when $\Omega = \pi$ (or $\Omega = -\pi$), i.e. $F = 1/2$ (or $F = -1/2$).



الفكرة x(n) = A cos(ωn + θ) is periodic only for rational f, frequencies 2π apart are identical, and the fastest oscillation is at ω = π (f = 1/2).

المصدر — من الـ PDFs بتاعتك

محاضرة 3

المحاضرة

LEC 3 · Sec. 1.5.2 · p. 9

1-Lectures/Lectures Fall 2026/Chapter 1/DSP- LEC- 3- Ch- 1.pdf

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Final 11-6-2023 | Final 7-6-2023 · Q4-a | Final 12-1-2019 | Final 2-1-2017 | Final 23-12-2017 | Midterm 26-7-2026 | Midterm 31-7-2023 | Midterm OLD | نظري فينال

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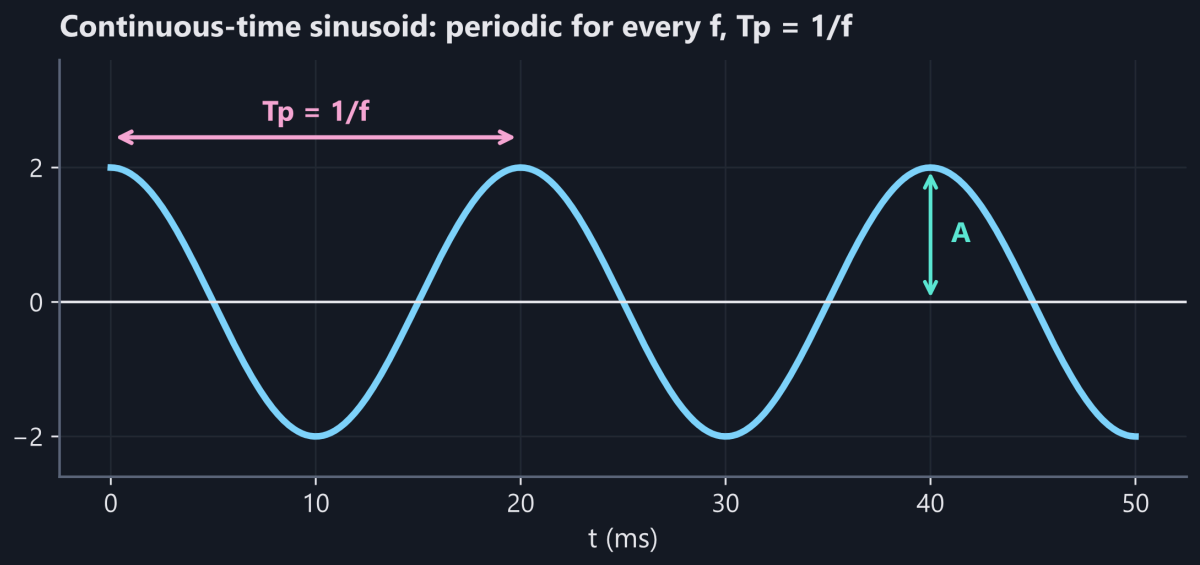
محاضرة 3

Q8 متكرر ●4×

In detail, what are the main characteristics of the continuous-time sinusoidal signals?

$x_a(t) = A\cos(\omega t + \theta) = A\cos(2\pi f t + \theta)$, where A is the amplitude, $\omega = 2\pi f$ the frequency in rad/s, f in Hz, and θ the phase in radians.

1. **(a)** For every fixed value of the frequency f, $x_a(t)$ is **periodic**: $x_a(t + T_p) = x_a(t)$, where $T_p = 1/f$ is the fundamental period.
2. **(b)** Continuous-time sinusoidal signals with distinct (different) frequencies are themselves **distinct**.
3. **(c)** Increasing the frequency f results in an **increase in the rate of oscillation** of the signal; f can be increased without limit.



الفكرة

$x_a(t) = A\cos(\Omega t + \theta)$ is always periodic ($T_p = 1/F$), distinct F give distinct signals, and a higher F means faster oscillation.

Key words periodic

المصدر — من الـ PDFs بتاعتك

المحاضرة

LEC 3 · Sec. 1.5.1 · p. 3

1-Lectures/Lectures Fall 2026/Chapter 1/DSP- LEC- 3- Ch- 1.pdf

جه في Final 20-1-2025 · Q3-a | Final 9-6-2026 · Q1-a | Quiz 1 Summer 26 · Q1-b | Midterm OLD

5-Exams/Final/ · 5-Exams/Midterm/ · 5-Exams/OLD Quiz/

×2 Q10

Show that a discrete-time sinusoid is periodic only if its frequency F is a rational number. How do you find its fundamental period?

By definition, x(n) is periodic with period N (N > 0) if and only if x(n + N) = x(n) for all n (1.15). For a sinusoid with frequency F_o to be periodic:

$$\cos[2\pi F_0(N + n) + \theta] = \cos(2\pi F_0 n + \theta)$$

This relation is true if and only if there exists an integer k such that $2\pi F_o N = 2k\pi$, or equivalently:

$$F_0 = k / N \quad (1.16)$$

So F_o must be rational. To find the fundamental period, cancel common factors so that k and N are relatively prime; then the fundamental period is N.

الفكرة x(n + N) = x(n) needs 2πfN = 2πk, so f = k/N must be rational; N (with k/N in lowest terms) is the fundamental period.

المصدر — من الـ PDFs بتاعتك

المحاضرة **LEC 3** · Sec. 1.5.2 · p. 9

1-Lectures/Lectures Fall 2026/Chapter 1/DSP- LEC- 3- Ch- 1.pdf

جه في Midterm 1 Model-2 · Q2-b | Notebook / Section

5-Exams/Midterm/ · DSP Student Notebook.pdf/

محاضرة 3

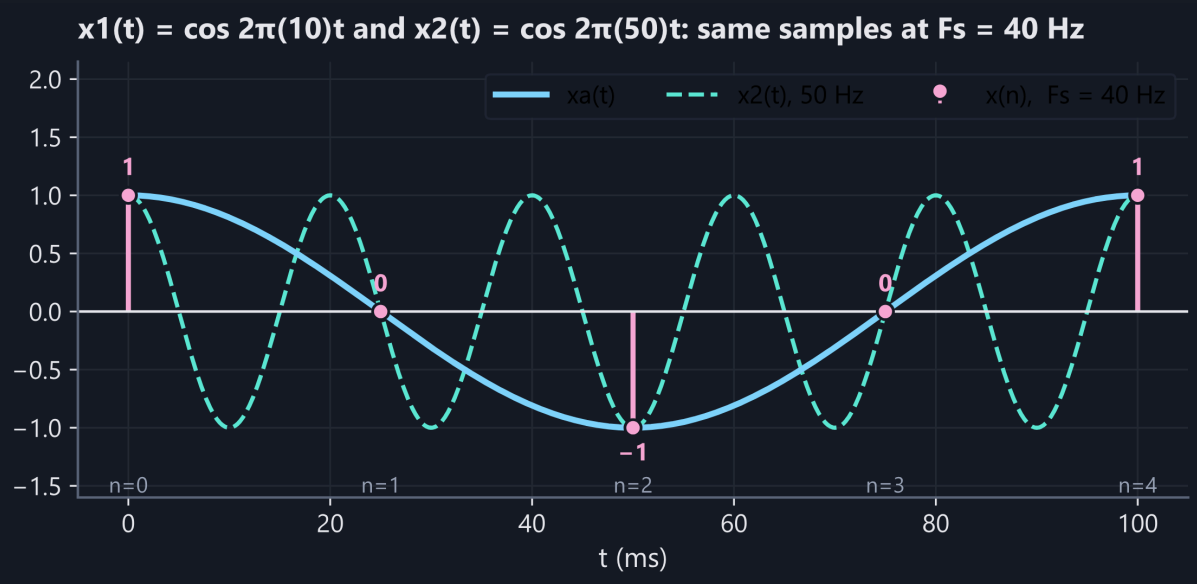
Q11 أقل تكرارًا

Prove that discrete-time sinusoids whose frequencies are separated by an integer multiple of 2π are identical. What is an alias?

$$\cos[(\Omega_0 + 2\pi)n + \theta] = \cos(\Omega_0 n + 2\pi n + \theta) = \cos(\Omega_0 n + \theta) \quad (1.17)$$

As a result, all sinusoidal sequences with $\Omega k = \Omega_o + 2k\pi$, $-\pi \le \Omega_o \le \pi$, are indistinguishable (identical).

Alias: any sequence from a sinusoid with $|\Omega| > \pi$ ($|F| > 1/2$) is identical to a sequence obtained from a sinusoid with $|\Omega| < \pi$; it is called an alias of the corresponding sinusoid.



الفكرة

cos((ω0 + 2πk)n) = cos(ω0 n) for integer n, so sinusoids 2πk apart are the same; any ωk = ω0 + 2πk is an alias of ω0.

Key words Alias

المصدر — من الـ PDFs بتاعتك

المحاضرة

LEC 3 · Sec. 1.5.2 (aliases) · p. 17

1-Lectures/Lectures Fall 2026/Chapter 1/DSP- LEC- 3- Ch- 1.pdf

محاضرة 3

Q12 أقل تكرارًا

What are harmonically related complex exponentials? Compare between the continuous-time and discrete-time cases.

Sets of periodic complex exponentials with fundamental frequencies that are multiples of a single positive frequency.

	Continuous time	Discrete time
Basic signals	$s_k(t) = e^{jk\Omega_o t}$, k = 0, ±1, ±2, ...	$s_k(n) = e^{j2\pi kn/N}$, k = 0, 1, ..., N − 1
Distinct members	all distinct (infinite)	only N distinct: $s_{k+N}(n) = s_k(n)$
Common period	$Tp = 1/F_o$	N samples
Fourier series	$xa(t) = \Sigma\, c_k\, e^{jk\Omega_o t}$	$x(n) = \Sigma_{k=0..N-1}\, c_k\, e^{j2\pi kn/N}$

الفكرة

s_k = e^{j2πkF0 t}: infinitely many distinct ones in continuous time, but only N distinct ones in discrete time (s_{k+N} = s_k).

المصدر — من الـ PDFs بتاعتك

المحاضرة

LEC 3 · Sec. 1.5.3 · p. 21

1-Lectures/Lectures Fall 2026/Chapter 1/DSP- LEC- 3- Ch- 1.pdf

محاضرة 3

★ Cheat sheet · محاضرة 3

افتكر

xa(t) = A cos(Ωt + θ) is always periodic (Tp = 1/F), distinct F give distinct signals, and a higher F means faster oscillation. **Q8**

x(n) = A cos(ωn + θ) is periodic only for rational f, frequencies 2π apart are identical, and the fastest oscillation is at ω = π (f = 1/2). **Q9**

x(n + N) = x(n) needs 2πfN = 2πk, so f = k/N must be rational; N (with k/N in lowest terms) is the fundamental period. **Q10**

cos((ω0 + 2πk)n) = cos(ω0 n) for integer n, so sinusoids 2πk apart are the same; any ωk = ω0 + 2πk is an alias of ω0. **Q11**

s_k = e^{j2πkF0 t}: infinitely many distinct ones in continuous time, but only N distinct ones in discrete time (s_{k+N} = s_k). **Q12**

المعادلات

$$\cos[(\Omega_0 + 2\pi)n + \theta] = \cos(\Omega_0 n + 2\pi n + \theta) = \cos(\Omega_0 n + \theta) \quad (1.17)$$

$$F_0 = k / N \quad (1.16)$$

SECTION 03

محاضرة 4

A/D and D/A Conversion

الأسئلة — الأكثر تكرارًا الأول

p. 23	Find the relationship between the frequency of the analog signal f, the frequency of...	Q15 ☐
p. 24	In A/D, explain in detail the quantization process of continuous-amplitude signals.	Q18 ☐
p. 25	In sampling process, derive the sampling condition Fs ≥ 2f max (avoiding aliasing).	Q16 ☐
p. 26	Explain (with a block diagram) the basic three operations performed in the A/D conve...	Q13 ☐
p. 28	In A/D, explain in detail the sampling process of analog signals.	Q14 ☐
p. 29	Check (✓) for the right statements and (×) for the wrong statements and correct the ...	Q22 ☐
p. 30	State and explain the sampling theorem. Define the Nyquist rate.	Q17 ☐
p. 31	In A/D conversion, explain in detail the coding process.	Q19 ☐
p. 31	Why is the sampling process invertible while the quantization process is not?	Q20 ☐
p. 32	Explain the D/A conversion process. Compare between ideal interpolation, zero-order ...	Q21 ☐

محاضرة 4

Q15 · متكرر ×8

Find the relationship between the frequency of the analog signal f, the frequency of the discrete-time signal F, and the sampling frequency Fs.

Consider xa(t) = A cos(2πft + θ). Sampling at Fs = 1/T (t = nT = n/Fs):

$$x(n) = xa(nT) = A\cos(2\pi fnT + \theta) = A\cos(2\pi nf/Fs + \theta) \quad (1.32)$$

Comparing with x(n) = A cos(2πFn + θ) (1.14):

$$F = f / Fs \quad (1.33) \qquad \text{or} \qquad \Omega = \omega T \quad (1.34)$$

الفكرة

t = nT gives F = f / Fs (and ω = Ω T): the digital frequency is the analog frequency divided by the sampling rate.

المصدر — من الـ PDFs بتاعتك

المحاضرة

LEC 4 · Sec. 1.6.1 · p. 8

1-Lectures/Lectures Fall 2026/Chapter 1/DSP- LEC- 4- Ch- 1.pdf

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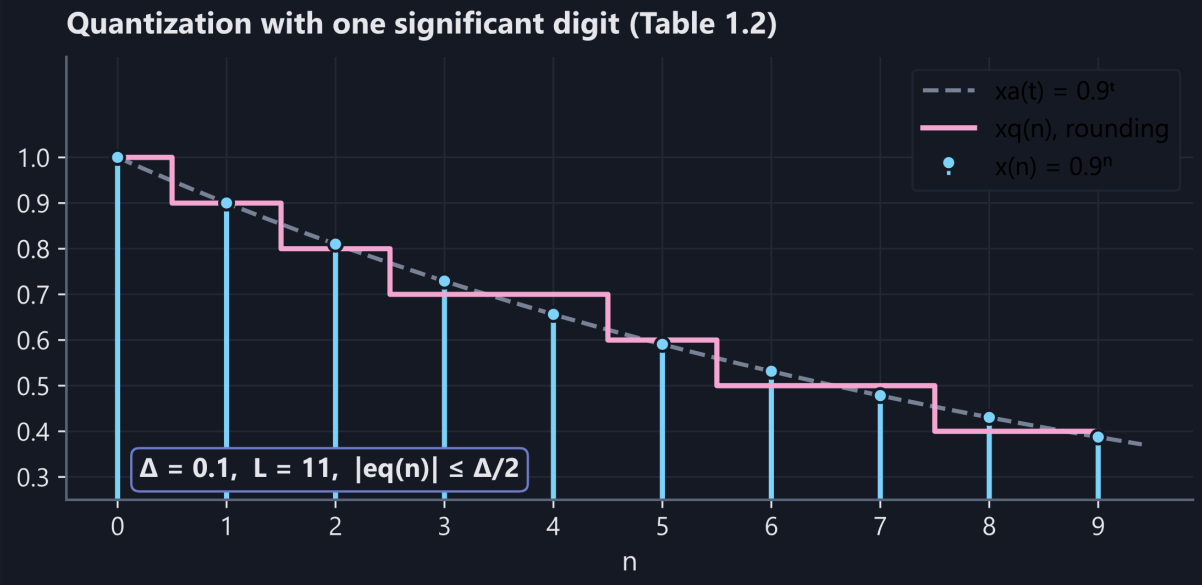
Final 18-1-2024 · Q1-a | Final 12-1-2019 · Q2-a | Final 2-1-2017 · Q2-a | Final 23-12-2017 · Q2-a | Final 28-12-2019 · Q1-a | Midterm 12-11-2025 · Q1-a | Midterm 2 · Q2-a | Final 8-9-2021

5-Exams/Final/ · 5-Exams/Midterm/

Q18 متكرر ●8×

In A/D, explain in detail the quantization process of continuous-amplitude signals.

- **Quantization:** the process of converting a discrete-time continuous-amplitude signal into a digital signal by expressing each sample value as a finite number of digits.
- **Quantization error:** eq(n) = xq(n) − x(n) (1.53), where xq(n) = Q[x(n)].
- The values allowed are the **quantization levels**; the distance between two successive levels is the **step size Δ**.
- In rounding: $-\Delta/2 \le eq(n) \le \Delta/2$ (1.54), and $\Delta = (x_{max} - x_{min})/(L - 1)$ (1.55).
- Increasing the number of levels L decreases Δ, so the quantization error decreases and the accuracy increases.



الفكرة Each sample is mapped to one of L levels with step Δ = (xmax − xmin)/(L − 1); the error eq(n) is at most Δ/2 when rounding.

Key words: Quantization, Quantization error

المصدر — من الـ PDFs بتاعتك

المحاضرة **LEC 4** · Sec. 1.6.3 · p. 23
1-Lectures/Lectures Fall 2026/Chapter 1/DSP- LEC- 4- Ch- 1.pdf

جه في
Final 7-6-2023 · Q1-a | Final 17-8-2026 · Q1-a | Final 4-3-2021 · Q1-a | Final 8-9-2021 · Q1-a | Midterm 2 · Q1-b | Quiz 1 · Section | Midterm OLD | نظري فينال
5-Exams/Final/ · 5-Exams/Midterm/ · 6-Extra/

Q16 متكرر ×6

In sampling process, derive the sampling condition $Fs \geq 2f_{max}$ (avoiding aliasing).

Since the discrete-time frequency must satisfy $-1/2 \leq F \leq 1/2$ and $F = f/Fs$:

$$-1/2 \leq f/Fs \leq 1/2 \rightarrow -Fs/2 \leq f \leq Fs/2 \quad (1.37)$$
$$f_{max} = Fs/2 \rightarrow \mathbf{Fs \geq 2f_{max}}$$

Thus to avoid the problem of aliasing, Fs is selected so that $Fs > 2f_{max}$ (1.47).

الفكرة
$|F| \leq 1/2$ with $F = f/Fs \rightarrow f \leq Fs/2$ for every component $\rightarrow Fs \geq 2\ fmax$; otherwise higher frequencies alias.

المصدر — من الـ PDFs بتاعتك

المحاضرة **LEC 4** · Sec. 1.6.1 · p. 8
1-Lectures/Lectures Fall 2026/Chapter 1/DSP- LEC- 4- Ch- 1.pdf

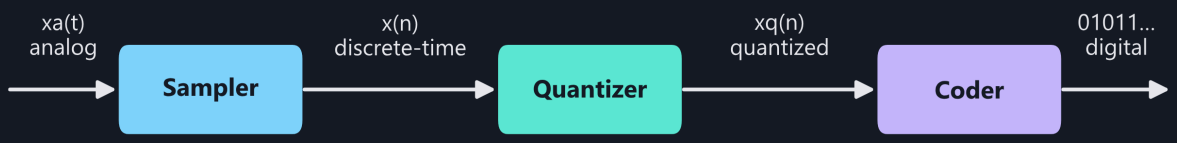
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Final 20-1-2025 · Q4-a | Final 27-6-2021 · Q1-a | Midterm 26-7-2026 · Q2-a | Midterm 31-7-2023 · Q2-a | Midterm 2 · Q1-a | Quiz 1 Summer 26 · Q1-a
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محاضرة 4

Q13 · متكرر · ×4

Explain (with a block diagram) the basic three operations performed in the A/D converter.

To process analog signals by digital means, it is first necessary to convert them into a sequence of numbers having finite precision. This is A/D conversion, a three-step process (Fig. 1.9):



1. **Sampling:** the conversion of a continuous-time signal into a discrete-time signal obtained by taking "samples" of the continuous-time signal at discrete-time instants. If xa(t) is the input to the sampler, the output is **x(n) = xa(nT)**, where T is the sampling interval.
2. **Quantization:** the conversion of a discrete-time continuous-valued signal into a discrete-time, discrete-valued (digital) signal. The value of each sample is selected from a finite set of possible values. The difference $x_q(n) - x(n)$ is the quantization error.
3. **Coding:** each discrete value $x_q(n)$ is represented by a **b-bit binary sequence**.

Sampling does not result in a loss of information if the signal bandwidth is finite (**invertible**), while quantization is a **noninvertible** process that results in signal distortion.

الفكرة A/D = sampler (discrete time) → quantizer (discrete values) → coder (b-bit binary words).

Key words: Sampling · Quantization · Coding

المصدر — من الـ PDFs بتاعتك

المحاضرة **LEC 4** · Sec. 1.6 · p. 3
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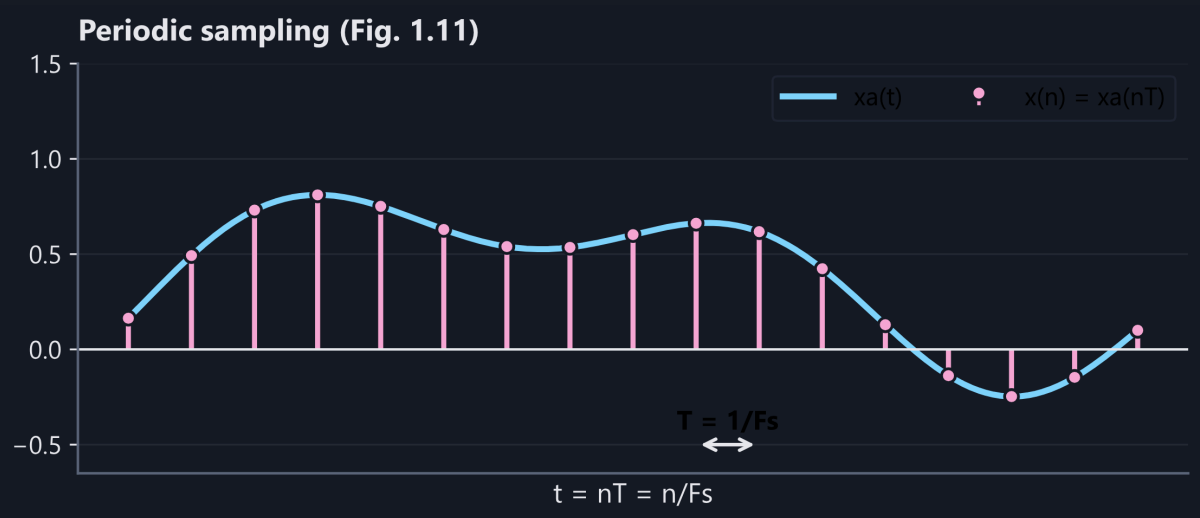
محاضرة 4

2-Section/Revision Sheet (summer)/ · 2-Section/Sheets/ · 5-Exams/Midterm/ · 6-Extra/

Q14 متكرر ×4

In A/D, explain in detail the sampling process of analog signals.

- We use periodic (uniform) sampling, described by the relation **x(n) = xa(nT)**, $-\infty < n < \infty$ (1.29), where x(n) is obtained by taking samples of xa(t) every T seconds (Fig. 1.11).
- The time interval T between successive samples is the **sampling period**; its reciprocal **Fs = 1/T** is the sampling rate (samples/s) or sampling frequency (Hz).
- Periodic sampling relates the time variables t and n: **t = nT = n/Fs** (1.30). As a consequence, the frequencies are related by F = f/Fs (1.33).
- Sampling does not result in a loss of information if the bandwidth is finite and Fs > $2f_{max}$.



الفكرة Periodic sampling takes x(n) = xa(nT), one sample every T = 1/Fs seconds.

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5-Exams/Midterm/

Q22 متكرر ●4×

Check (✓) for the right statements and (×) for the wrong statements and correct the mistakes.

1. In quantization process, when the quantization step (Δ) increases, the accuracy of the quantizer increases. **(×)**
   Correct: when Δ **decreases**, the accuracy increases.
2. Nyquist frequency (F_N) is the minimum sampling frequency. **(✓)**
3. To reconstruct x(t) from x(n), the sampling rate Fs must be less than the frequency of x(t), f_o. **(×)**
   Correct: Fs must be **greater than $2f_o$**.

الفكرة

Smaller Δ → more accurate (× as written); Nyquist rate = minimum sampling rate (✓); reconstruction needs Fs > $2f_o$ (× as written).

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المحاضرة

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5-Exams/Final/ · 5-Exams/OLD Quiz/

×3 Q17

State and explain the sampling theorem. Define the Nyquist rate.

- To select Fs we must have information about the frequency content of the signal, e.g. f_{max} = 3000 Hz for speech and 5 MHz for television signals. The signal may be passed through a filter that severely attenuates components above f_{max}.
- To avoid aliasing, Fs is selected so that **$Fs > 2f_{max}$** (1.47). Then every component $|f_i| < f_{max}$ is mapped into $-1/2 \le F_i = f_i/Fs \le 1/2$ (1.48).

Sampling theorem: if the highest frequency contained in an analog signal xa(t) is f_{max} = B and the signal is sampled at a rate $Fs > 2f_{max} = 2B$, then xa(t) can be exactly recovered from its sample values using the interpolation function:

$$g(t) = \frac{\sin(2\pi Bt)}{2\pi Bt} \quad (1.50) \qquad x_a(t) = \sum x_a(n/Fs)\, g(t - n/Fs) \quad (1.51)$$

- The sampling rate **$f_N = 2B = 2f_{max}$** is called the **Nyquist rate**.
- Because of the complexity and the infinite number of samples required, these reconstruction formulas are primarily of theoretical interest.

الفكرة A band-limited signal (fmax) is recovered exactly from its samples if Fs > 2 fmax; FN = 2 fmax is the Nyquist rate.

Key words: Sampling theorem

المصدر — من الـ PDFs بتاعتك

المحاضرة
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1-Lectures/Lectures Fall 2026/Chapter 1/DSP- LEC- 4- Ch- 1.pdf

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Sheet 1 · 1.5 | نظري الميد | نظري Eng.Khalaf
2-Section/Eng.Khalaf/ · 2-Section/Sheets/ · 6-Extra/

×1 Q19

In A/D conversion, explain in detail the coding process.

- The coding process assigns a unique binary number to each quantization level.
- With L levels we need at least L binary numbers; with b bits we can create 2^b numbers.

$$2^b \geq L \quad \rightarrow \quad b \geq \log_2 L$$

الفكرة Each quantized level gets a unique b-bit binary number, with 2^b ≥ L.

المصدر — من الـ PDFs بتاعتك

المحاضرة **LEC 4** · Sec. 1.6.4 · p. 32
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7-Books (References)/Extract Question From Book/

أقل تكرارًا Q20

Why is the sampling process invertible while the quantization process is not?

- **Sampling** does not result in a loss of information, nor introduce distortion, if the signal bandwidth is finite → **invertible (reversible)**.
- **Quantization** is a **noninvertible (irreversible)** process that results in signal distortion.

الفكرة Sampling a band-limited signal loses nothing (invertible); quantization throws away the exact value, so the error cannot be undone.

المصدر — من الـ PDFs بتاعتك

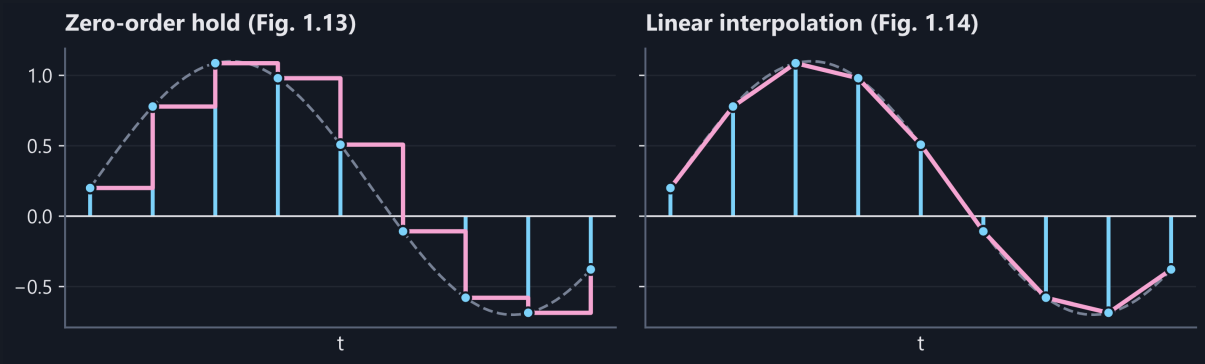
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Q21 أقل تكرارًا

Explain the D/A conversion process. Compare between ideal interpolation, zero-order hold and linear interpolation.

- The task of a D/A converter is to **interpolate between samples**. The ideal interpolation (sampling theorem) is too complicated and impractical.
- **Zero-order hold:** the simplest; holds constant the value of one sample until the next one is received.
- **Linear interpolation:** connects successive samples with straight-line segments (better).



الفكرة D/A connects the samples: ideal interpolation is impractical, zero-order hold keeps each value constant, linear interpolation joins samples with straight lines.

Key words Zero-order hold Linear interpolation

المصدر — من الـ PDFs بتاعتك

المحاضرة

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محاضرة 4

★ Cheat sheet · محاضرة 4

افتكر

A/D = sampler (discrete time) → quantizer (discrete values) → coder (b-bit binary words). **Q13**

Periodic sampling takes x(n) = xa(nT), one sample every T = 1/Fs seconds. **Q14**

t = nT gives F = f / Fs (and ω = Ω T): the digital frequency is the analog frequency divided by the sampling rate. **Q15**

|F| ≤ 1/2 with F = f/Fs → f ≤ Fs/2 for every component → Fs ≥ 2 fmax; otherwise higher frequencies alias. **Q16**

A band-limited signal (fmax) is recovered exactly from its samples if Fs > 2 fmax; FN = 2 fmax is the Nyquist rate. **Q17**

Each sample is mapped to one of L levels with step Δ = (xmax − xmin)/(L − 1); the error eq(n) is at most Δ/2 when rounding. **Q18**

Each quantized level gets a unique b-bit binary number, with 2^b ≥ L. **Q19**

Sampling a band-limited signal loses nothing (invertible); quantization throws away the exact value, so the error cannot be undone. **Q20**

D/A connects the samples: ideal interpolation is impractical, zero-order hold keeps each value constant, linear interpolation joins samples with straight lines. **Q21**

Smaller Δ → more accurate (× as written); Nyquist rate = minimum sampling rate (✓); reconstruction needs Fs > $2f_o$ (× as written). **Q22**

المعادلات

$$F = f / Fs \quad (1.33) \qquad \text{or} \qquad \Omega = \omega T \quad (1.34)$$

$$x(n) = xa(nT) = A\cos(2\pi fnT + \theta) = A\cos(2\pi nf/Fs + \theta) \quad (1.32)$$

$$g(t) = \sin(2\pi Bt) / (2\pi Bt) \quad (1.50)$$

$$xa(t) = \Sigma\, xa(n/Fs)\, g(t - n/Fs) \quad (1.51)$$

$$-1/2 \le f/Fs \le 1/2 \;\rightarrow\; -Fs/2 \le f \le Fs/2 \quad (1.37)$$

$$f_{max} = Fs/2 \;\rightarrow\; \mathbf{Fs \ge 2f_{max}}$$